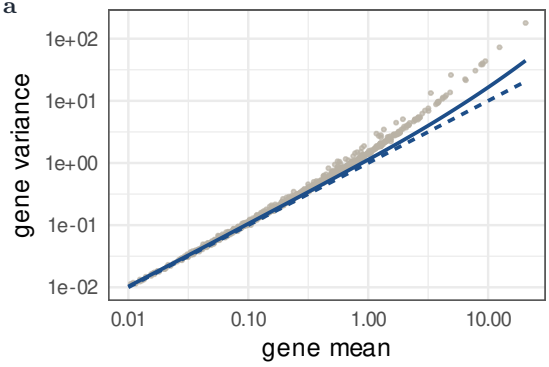
Negative-binomial variance

$$\mathrm{Var}(x_{gc}) \;=\; m_{gc} \;+\; m_{gc}^2/\psi_g$$

Depth as an offset, no cell type ($M_0$)

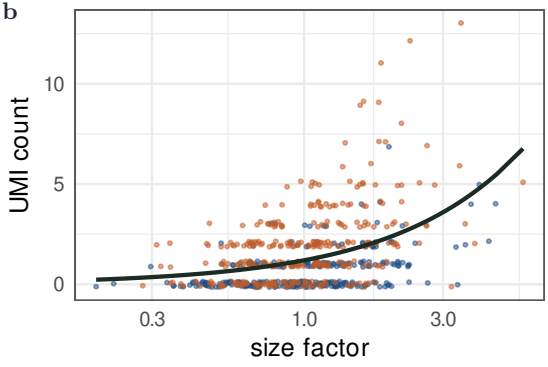
$$\log m_{gc} \;=\; \beta_{0g} \;+\; \log s_c$$

**a**




NB, smoothed dispersion  Poisson

**b**




type 1  type 2

one $\hat{\psi}_g$ per gene

Regularise across genes

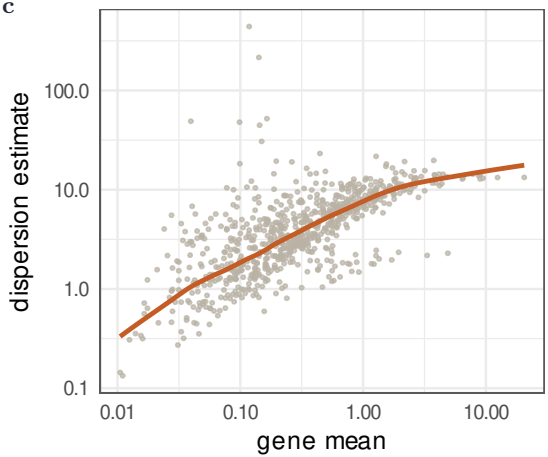
$$\hat{\psi}_g \;\rightarrow\; \tilde{\psi}_g \;=\; \mathrm{smooth}\{\hat{\psi}_g \;|\; \log \bar{x}_g\}$$

fitted $\hat{m}_{gc}$

Pearson residuals, clipped at $\pm\sqrt{C/30}$

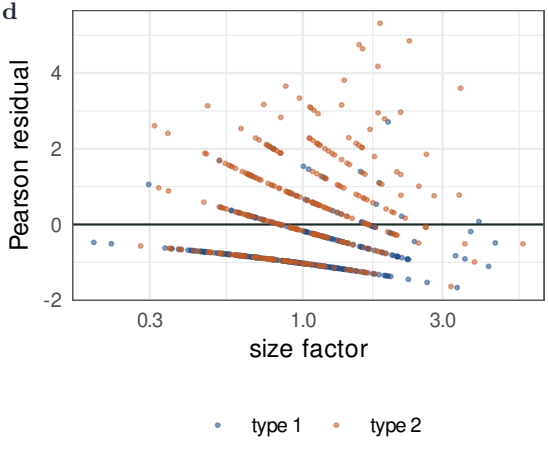
$$r_{gc} \;=\; (x_{gc} - \hat{m}_{gc})/\sqrt{\hat{m}_{gc} + \hat{m}_{gc}^2/\tilde{\psi}_g}$$

**c**




$\tilde{\psi}_g$

**d**




type 1  type 2